Yufeng (Alex) Xia

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Profile

- MSc by Research student at the Institute for Computing Systems Architecture (ICSA), University of Edinburgh, supervised by Prof. Mahesh K. Marina. My interest lies at the intersection of **wireless systems, systems for AI, and large-scale emulation**. My work spans **AI-RAN**, where live radio processing and data-driven workloads share the same infrastructure under real-time constraints, as well as **emulation and digital twin systems**, the platforms on which RAN and edge-AI algorithms are developed and validated at scale.

  I build the layer between the radio and the model, making live network behaviour something models can be trained and validated against, and idle radio compute something they can run on. In practice this means systems engineering in C/C++ and Python on Linux and GPU platforms, with research at ACM MobiCom and EdgeSys (Best Paper), and a strong interest in where these ideas go in 6G. **Available for a full-time internship from September 2026.**

Education

- **University of Edinburgh — MSc by Research, Computer Science** Edinburgh, UK
  Supervised by Prof. Mahesh K. Marina. Thesis: Flux-DT: An Elastic, Preemption-Tolerant Digital Twin Runtime for AI-RAN *Sep 2025 – Sep 2026 (expected)*

- **University of Edinburgh — BSc Computer Science** Edinburgh, UK
  *Grade: **First Class Honours*** *Sep 2021 – May 2025*
  - **Relevant courses:** Machine Learning Systems, Machine Learning Practical, Computer Communications and Networks.

Technical Skills

- **Languages** C, C++, Python, Java, Bash.
- **Wireless & O-RAN** OpenAirInterface, Open5GS, RF simulator, 5G NR PHY/MAC experimentation, ray-traced channel modelling (Sionna RT), RAN emulation and network digital twins.
- **AI/ML & Systems** PyTorch, distributed foundation-model training runs, near-real-time resource-allocation policies and heuristics, CUDA GPU programming and profiling, workload scheduling, large-scale system benchmarking.
- **Infrastructure** Linux, Docker, Kubernetes, Git, Microsoft Azure, CloudLab.

Research and Project Experience

- **Flux-DT — Elastic, Preemption-Tolerant Digital Twin Runtime for AI-RAN** **MSc by Research Thesis**
  University of Edinburgh — sole author *Sep 2025 – Present*
  - **Sharing RAN compute with a stateful workload**: Existing AI-RAN compute-sharing designs assume the co-located workload is stateless and disposable, whereas a radio digital twin is neither: it holds scene state that must survive interruption. Built a runtime that lets such a twin run on bursty spare compute in the radio path of a live OpenAirInterface 5G stack, without violating live RAN priorities or corrupting its own state.
  - **Near-real-time control policy**: Designed and implemented the control loop that responds to radio-driven resource pressure with a graded set of actions, on the near-real-time timescale the radio imposes. The live RAN retains priority throughout, while the twin preserves its state.
  - **Multi-cell fidelity placement algorithm**: Formulated fidelity placement as a joint objective across co-channel cells and **designed a greedy heuristic** that decides which tiles are worth solving under interference; the same ranking determines which work is shed first when spare compute is reclaimed mid-solve.

- **EmuRAN — Scalable, Cost-Effective Cloud-based RAN Emulation System** MobiCom 2026
  *Research Assistant; **owned the end-to-end evaluation*** *Conditionally Accepted*
  - **Public cloud operation**: Deployed and operated the emulator on public cloud infrastructure, emulating over 10,000 mobile devices, two orders of magnitude larger in scale than existing software solutions.

  - **Fidelity and scalability validation**: Designed and ran the evaluation campaign: fidelity against a real RAN (contention-based random access timelines, iperf3 throughput comparison, per-slot completion-time distributions) and scalability (time-dilation behaviour against node count and under constrained compute).

- **Weaver — Foundation Model Training over AI-RAN Compute Infrastructure** Under Submission
  Research Assistant; spare-compute controller evaluation and large-scale experiments *2024 – 2026*
  - **RAN-centric spare compute control**: Evaluated the controller that decides how much accelerator capacity an AI workload may harvest from shared RAN infrastructure, a near-real-time resource-allocation problem on GPU-accelerated RAN hardware. On a multi-site testbed it harvests up to **83%** of the available spare compute without degrading live radio performance.
  - **PHY experiments and large-scale training runs**: Ran PHY-layer experiments and GPU profiling to characterise the interference between training workloads and radio processing, and carried out the large-scale foundation-model training runs used in the evaluation.

- **CoSenseAQ — LLVM Auto-Quantization from Sensor Specifications** Under Submission
  Research Assistant, ICSA; first author *Apr 2024 – May 2025*
  - **Sensor specifications as compiler input**: Built an LLVM auto-quantization framework that drives optimisation from sensor specifications, information a compiler cannot recover from source alone, reducing code size and execution time on embedded targets.
  - **Embedded evaluation**: Evaluated the framework on embedded ARM devices, measuring power, performance and area improvements; low-level C/C++ and cross-compilation toolchain work throughout.

- **Morphling — Emulator for Distributed Machine Learning at the Edge** EdgeSys 2026
  *Research Assistant; **Best Paper Award*** *Oct 2025 – Mar 2026*
  - **Measurement-driven edge emulation**: Contributed to a measurement-driven emulator for evaluating distributed ML training on heterogeneous edge devices, supporting up to 1,800 emulated devices on a single GPU server with low latency and thermal modelling error.

Industry Experience

- **Senior Tech — Back-end Development Intern** Suzhou, China
  Production backend APIs in a commercial engineering team; live platform migration to Spring Boot, code review and release testing. Java, MySQL, Redis, Kafka. *Jun 2023 – Sep 2023*

Selected Publications

EmuRAN: Scalable and Cost-Effective Cloud-based RAN Emulation System *MobiCom'26 (cond. accepted)*
*Ujjwal Pawar, Andrew E. Ferguson, **Yufeng Xia**, Xenofon Foukas, Mahesh Marina, Bozidar Radunovic*

Weaver: Foundation Model Training over AI-RAN Compute Infrastructure *under submission*
*Leyang Xue, Tianxin Wang, Xin Zhe Khooi, Jiaxun Yang, Dheeraj Mahendiran, **Yufeng Xia**, Mun Choon Chan, Myungjin Lee, Mahesh Marina*

Morphling: Emulator for Distributed Machine Learning at the Edge *EdgeSys'26 (Best Paper)*
*Leyang Xue, **Yufeng Xia**, Ismaeel Bashir, Eren Mendi, Jiaxun Yang, Myungjin Lee, Mahesh Marina*

Wasp: Foundation Model Training System for the Edge *SEC'26*
*Leyang Xue, **Yufeng Xia**, Myungjin Lee, Mahesh Marina*

CoSenseAQ: Compiler Optimizations using Sensor Technical Specifications with Automatic Quantization *under submission*
***Yufeng Xia**, Pei Mu, Antonio Barbalace*

Awards, Languages and Service

- **Award Best Paper**, EdgeSys 2026 (co-located with ACM MobiSys 2026), for *Morphling.*
- **Languages** Mandarin (native), English (fluent).
- **Service** Student volunteer, EuroSys 2026 (on-site conference operations).